

MTH205 Analysis in one variable

Tutorial 3

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- (1) Show that the function $f(x) = |x|$ for all $x \in \mathbb{R}$ is continuous on $\mathbb{R}$.

Solution: Let $a \in \mathbb{R}$. Then

$$|f(x) - f(a)| = ||x| - |a|| \leq |x - a|, \text{ for every } x \in \mathbb{R}.$$

For $\epsilon > 0$ choose $\delta = \epsilon$, it follows from the above equation that $|f(x) - f(a)| < \epsilon$ whenever $|x - a| < \delta$.

- (2) Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous such that $f(r) = 0$ for every $r \in \mathbb{Q}$ then prove that $f(x) = 0, \forall x \in \mathbb{R}$.

Solution: Let $x \in \mathbb{R}$. As the set $\mathbb{Q}$ of rational numbers is dense in $\mathbb{R}$, we can choose a sequence $\{r_n\}$ in $\mathbb{Q}$ such that $\{r_n\} \rightarrow x$ as $n \rightarrow \infty$. Since f is a continuous function, we have $\{f(r_n)\} \rightarrow f(x)$ as $n \rightarrow \infty$. On the other hand, the sequence $\{f(r_n)\} = \{0, 0, 0, \dots\} \rightarrow 0$ as $n \rightarrow \infty$. Using the fact that the limit of a convergent sequence is unique, we obtain $f(x) = 0$. Since $x \in \mathbb{R}$ was chosen arbitrarily, we conclude that $f(x) = 0$, for all $x \in \mathbb{R}$.

- (3) If $x \in \mathbb{R}$ we define $\lfloor x \rfloor$ to be the greatest integer $n \in \mathbb{Z}$ with $n \leq x$. Determine the points of continuity of the following functions defined on $\mathbb{R}$:

(a) $f(x) = \lfloor x \rfloor$

(c) $h(x) = \lfloor \sin(x) \rfloor$

(b) $g(x) = x \lfloor x \rfloor$

(d) $\varphi(x) = \lfloor \frac{1}{x} \rfloor$ for $x \neq 0$.

Solution: (a) Here the function $f(x)$ is also known as the “floor function”. For example, $f(1.8) = \lfloor 1.8 \rfloor = 1$. If $x \in \mathbb{Z}$ then $f(x) = x$ and the sequence $\{x - \frac{1}{n}\}$ is such that

$$\{x - \frac{1}{n}\} \rightarrow x \text{ but } \{f(x - \frac{1}{n})\} = \{(x-1), (x-1), (x-1), \dots\} \rightarrow (x-1) \neq f(x),$$

as $n \rightarrow \infty$. This shows that f is discontinuous at every integer. Now, it is easy to see that f is continuous at every point in $\mathbb{R} \setminus \mathbb{Z}$.

(b) The function $g(x) = x \lfloor x \rfloor$ is continuous at x if and only if $x \neq \pm 1, \pm 2, \pm 3, \dots$.

(c) The function $h(x) = \lfloor \sin(x) \rfloor$ is continuous at x if and only if $\sin(x) \neq 0, 1$.

(c) The function $\varphi(x) = \lfloor \frac{1}{x} \rfloor$ is continuous at x if and only if $x \neq 0, \pm \frac{1}{2}, \pm \frac{1}{3}, \dots$

- (4) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Then prove that $S = \{x \in \mathbb{R} / f(x) = 0\}$ is a closed subset of $\mathbb{R}$.

Solution: First, note that $\{0\}$ is a closed subset of $\mathbb{R}$. Since f is a continuous function, from one of the equivalent conditions, we conclude that $S = f^{-1}(\{0\})$ is closed in $\mathbb{R}$.

(5) Prove that the function $F: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$F(x) = \begin{cases} 0, & x = 0 \\ x \sin(\frac{1}{x}), & x \neq 0 \end{cases}$$

is continuous at $x = 0$.

Solution: We prove the continuity of F at $x = 0$ using the sequential criterion. Suppose $\{x_n\}$ is a sequence in $\mathbb{R} \setminus \{0\}$ converges to 0, by using the fact that $-1 \leq \sin(\theta) \leq 1$ for all $\theta \in \mathbb{R}$ we have the inequality,

$$-x_n \leq x_n \sin\left(\frac{1}{x_n}\right) \leq x_n, \quad \text{for all } n \in \mathbb{N}.$$

Since $\{x_n\}$ converges to 0, by the Sandwich theorem of sequences, it follows that

$$\{F(x_n)\} = \left\{x_n \sin\left(\frac{1}{x_n}\right)\right\} \rightarrow 0 = F(0),$$

as $n \rightarrow \infty$. Therefore, F is continuous at $x = 0$.